



EAST WEST UNIVERSITY

Department of Computer Science & Engineering

Graph-Based sEMG Gesture Recognition Using NinaPro DB2

TASK 2 REPORT: Baseline Models and Proposed ST-EGAT

Submitted To

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1. Task Overview

Task 2 asks a simple question: after building a strong feature-based baseline, does explicitly modeling the 12 electrodes as a graph add useful information? We first converted each 200 ms window into a compact engineered representation containing 15 descriptors per electrode plus pairwise electrode correlations. All preprocessing steps - imputation, variance filtering, ANOVA feature selection, scaling and class weighting - were fitted only on the training repetitions. Seven conventional models were then evaluated, with XGBoost emerging as the strongest baseline at 77.69% accuracy and 77.39% Macro-F1. We next trained the proposed ST-EGAT, which combines raw temporal features, handcrafted node descriptors, a sample-specific electrode graph and the same 200 selected global features used by the baseline pipeline. The proposed model reached 81.16% accuracy and 80.98% Macro-F1, giving a clear improvement over XGBoost on the same repetition-wise split.

2. Data Representation and Preprocessing

Component	Implementation
Raw window	12 channels $\times$ 400 samples (200 ms at 2000 Hz)
Per-electrode features	15 time/frequency descriptors $\times$ 12 electrodes = 180 features
Pairwise correlations	66 electrode-pair correlation features
Initial engineered feature count	246
After variance filter	234
Selected global features	Top 200 ANOVA features, training-only
Training matrix	121,173 $\times$ 200
Test matrix	60,261 $\times$ 200
Class imbalance ratio	1.816:1
Imbalance handling	Training-only class weights, range 0.7437–1.3507
Scaling	Median imputation + variance filtering + ANOVA selection + StandardScaler

The feature ranking also gives a useful sanity check on the engineered representation. Several of the highest-ranked ANOVA features come from Channels 11 and 4, including WAMP, RMS, DASDV, IEMG and MAV descriptors. In other words, the model-ready feature set is not driven by a single statistic or a single electrode; it combines amplitude, variation and complexity information from multiple channels.

3. Baseline Evaluation

Model	Accuracy	Macro-F1	ROC-AUC	PR-AUC	Train time
XGBoost	77.69%	77.39%	98.25%	85.99%	308.38 s
Random Forest	77.24%	76.99%	97.74%	82.98%	598.38 s
Gradient Boosting	75.89%	75.62%	97.96%	83.91%	441.25 s
MLP	70.32%	70.10%	97.50%	79.93%	42.93 s
Decision Tree	46.77%	46.52%	73.09%	29.35%	70.06 s
Logistic Regression	34.85%	35.28%	81.03%	32.51%	84.55 s
Linear SVM	35.15%	33.15%	82.46%	33.87%	994.70 s

XGBoost was selected as the baseline to beat because it produced the highest Macro-F1. Random Forest was very close, which confirms that tree ensembles are well suited to the non-linear structure of the engineered sEMG features. Logistic Regression and Linear SVM were much weaker, suggesting that the 17-class boundary cannot be captured well by a simple linear separator.

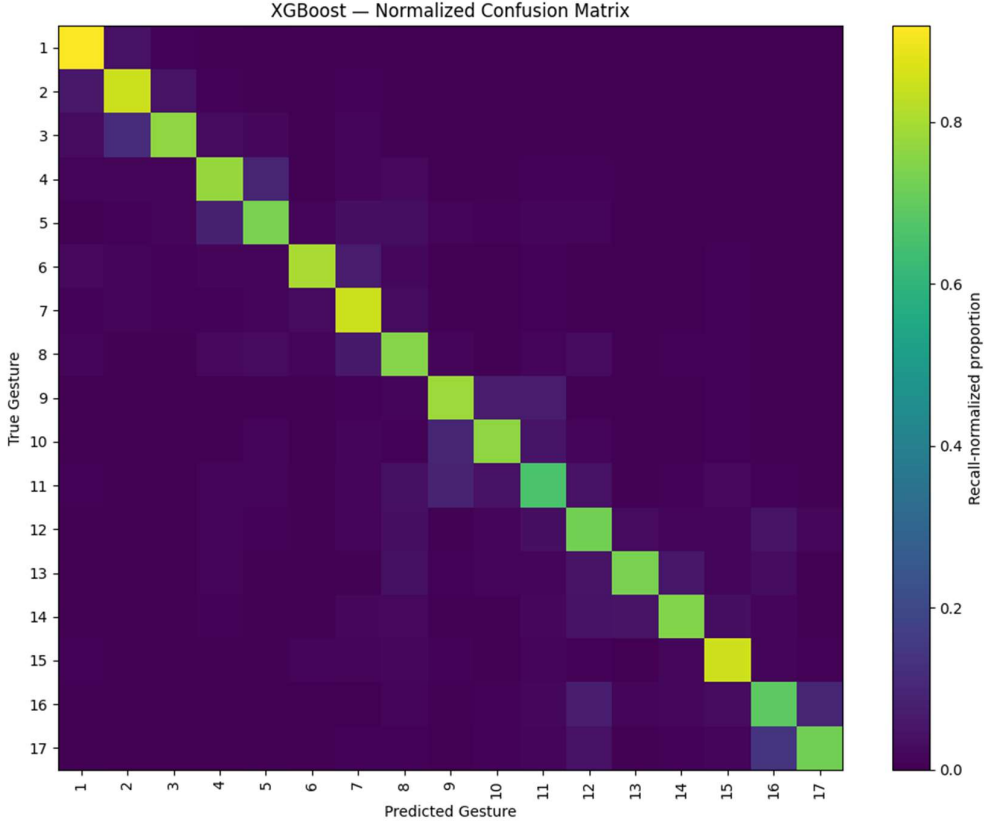


Figure 1. Normalized confusion matrix for the strongest baseline (XGBoost).

The XGBoost confusion matrix is strongly diagonal across most gestures, but the remaining off-diagonal errors leave room for a model that can use electrode relationships as well as global features.

4. Proposed ST-EGAT Model

Our proposed model keeps the physical meaning of the sEMG setup. Each electrode is represented as a graph node. For every 200 ms window, a shared temporal CNN learns waveform information from each channel, while 15 handcrafted descriptors provide a compact summary of that electrode's signal. The strongest sample-specific correlations define a sparse graph, and three edge-aware multi-head attention blocks exchange information between connected electrodes. The graph representation is then combined with a residual MLP branch built from the same 200 selected global features used in the baseline pipeline. This design lets the model use three complementary views of the same window: raw temporal shape, electrode-to-electrode relationships and strong global tabular features.

Stage	Role
Input	Raw 12×400 window plus 15 handcrafted features per electrode
Temporal encoder	Shared 1D-CNN extracts a learned embedding for each electrode
Node fusion	Temporal embedding and handcrafted descriptors are fused into node representations
Graph	12 electrode nodes; sample-specific top-k=5 correlation connectivity; signed Pearson weights
Graph learner	Three edge-aware multi-head graph-attention blocks in the implemented model
Readout	Learned attention pooling + mean pooling + max pooling
Residual global branch	MLP over the same 200 selected baseline features
Fusion / output	Graph and global representations are fused and classified into 17 gestures

4.1 Training configuration

Setting	Value
Epochs	55
Optimizer	AdamW
Learning rate	1.0×10^{-3}
Weight decay	3×10^{-4}
Schedule	OneCycleLR with cosine annealing
Loss	Class-weighted cross-entropy, label smoothing 0.02
Auxiliary loss weight	0.30 for graph/global expert supervision
Numerical efficiency	Automatic mixed precision (AMP)
Weight averaging	EMA decay 0.98 from epoch 8, update every 4 steps

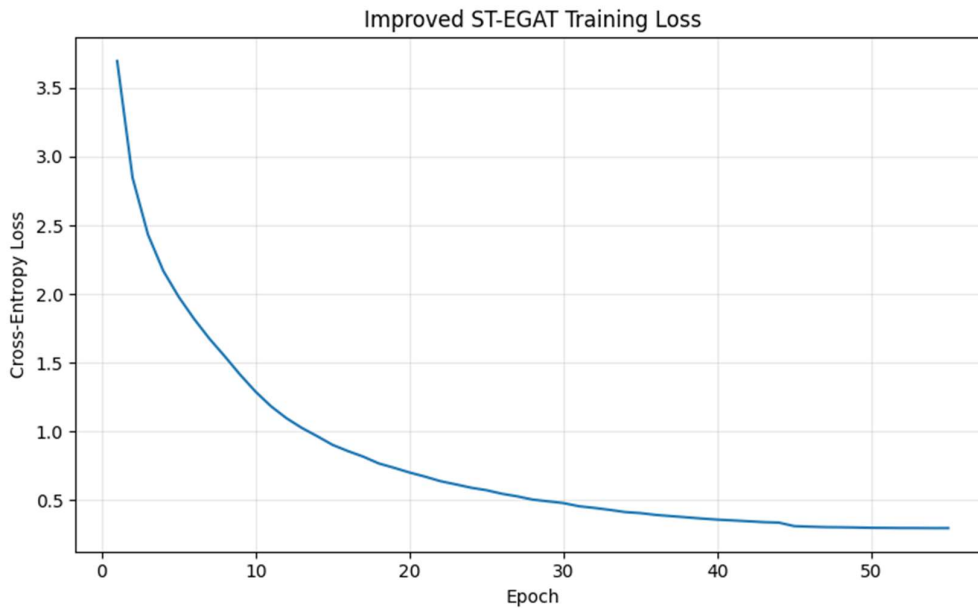


Figure 2. ST-EGAT training loss across 55 epochs.

Training loss falls smoothly over the 55 epochs without abrupt instability, indicating stable optimization on the training repetitions.

5. First GNN Results

Model	Accuracy	Macro-F1	ROC-AUC	PR-AUC	Train time
XGBoost baseline	77.69%	77.39%	98.25%	85.99%	308.38 s
Proposed ST-EGAT	81.16%	80.98%	97.66%	86.93%	3495.64 s

Primary result. ST-EGAT improves Accuracy by +3.46 percentage points and Macro-F1 by +3.59 percentage points over XGBoost. PR-AUC also increases from 85.99% to 86.93%, while XGBoost retains a slightly higher macro ROC-AUC (98.25% vs 97.66%).

The first GNN result is encouraging but also nuanced. ST-EGAT improves Accuracy by 3.46 percentage points and Macro-F1 by 3.59 points over XGBoost. PR-AUC also increases slightly, while XGBoost keeps the higher macro ROC-AUC. We therefore describe the result as a meaningful classification improvement, especially in Macro-F1, rather than claiming that the GNN dominates every metric.

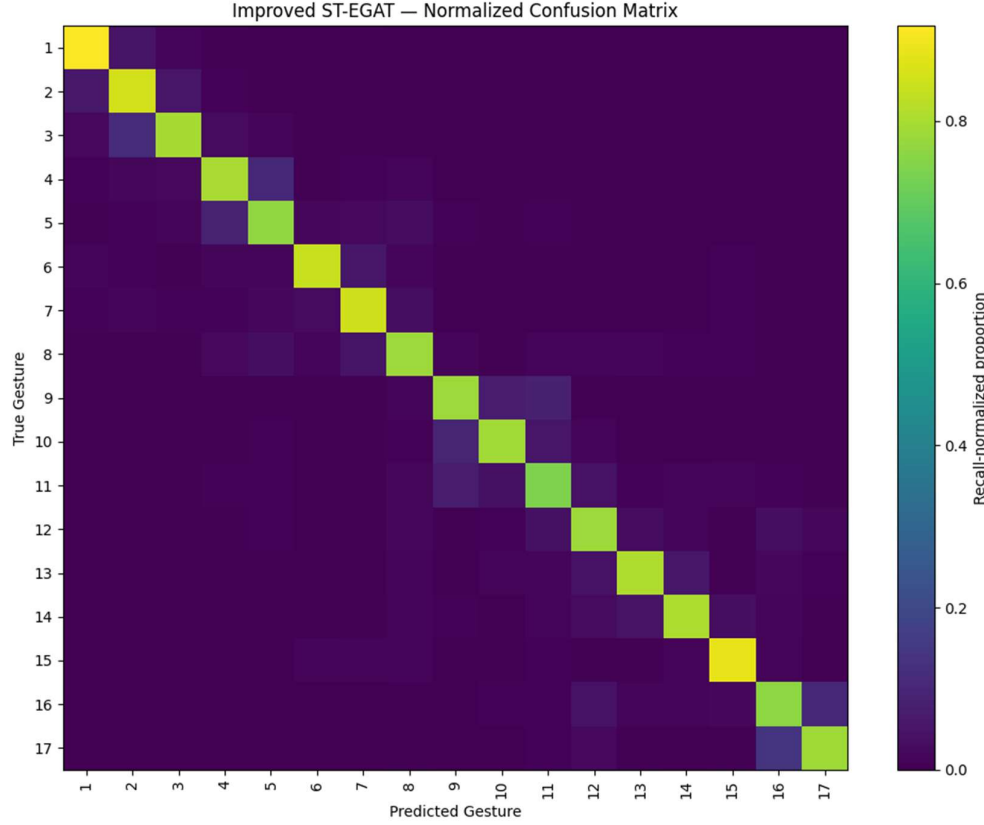


Figure 3. Normalized confusion matrix for the proposed ST-EGAT.

Most gestures remain strongly concentrated on the diagonal. Gesture 1 has the highest recall, while Gesture 11 is the clearest class-level weakness and becomes a natural target for Task 3.

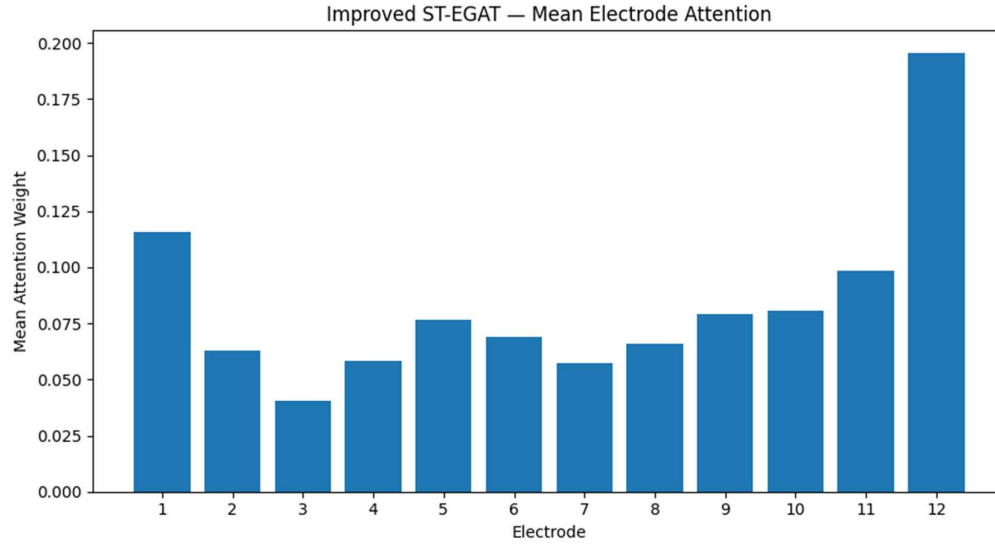


Figure 4. Average electrode attention at graph readout.

The graph readout is not uniform across electrodes. Channel 12 receives the largest average attention, which motivates a closer sample-level explanation in Task 3.

6. How the Proposed Model Works

1. A 200 ms sEMG window is first kept as 12 separate electrode signals instead of being flattened immediately.
2. The same temporal CNN is applied to every electrode so the model can learn waveform shape while using a shared representation rule across channels.
3. Handcrafted descriptors are added to the learned temporal representation because they summarize amplitude, variation, frequency and signal complexity in a compact form.

4. Electrodes are connected according to the strongest correlations within the current window, so message passing is guided by the relationships that are active for that sample.
5. Graph-attention layers learn how much information to exchange among connected electrodes rather than treating every connection equally.
6. Attention, mean and max readouts summarize the full electrode graph, while a separate MLP preserves the strong 200-feature tabular context learned by the baseline pipeline.
7. The two branches are fused to predict one of the 17 Exercise B gestures.

Why the graph can help: XGBoost is strong on engineered global features, but it does not directly represent the 12-electrode relationship structure or raw waveform morphology. ST-EGAT keeps the successful global feature branch and adds those two sources of information. This gives the model a chance to learn both what each electrode is doing and how electrodes are interacting within the same window.

7. Conclusion

Task 2 establishes a fair baseline-to-GNN comparison under one consistent protocol. XGBoost remains a strong benchmark, but the first ST-EGAT raises Macro-F1 to 80.98% while preserving a realistic view of the trade-offs across ROC-AUC and PR-AUC. The next task is not simply to add more complexity. Instead, Task 3 tests which graph choices actually matter, whether the gain is stable across grouped folds, and whether the final model can be explained at the feature and electrode level.